Pragya Goyal pragyathehuman@gmail.com

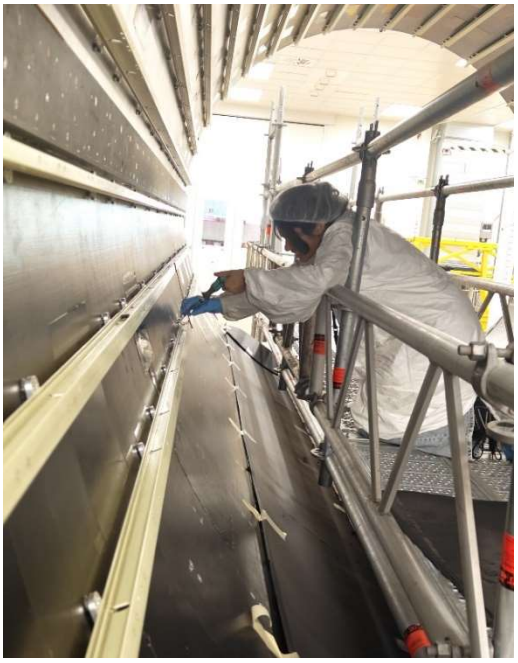
CERN Summer Internship | June 2026

CMS Upgrade: Detector Laboratory Work

I worked inside a cleanroom on the CMS detector upgrade, gaining hands-on experience with the construction and preparation of particle-detector components. The work was part of the broader upgrades being carried out to the Large Hadron Collider (LHC), which will allow it to operate at much higher luminosity and produce significantly more proton-proton collisions. This will enable the CMS experiment to collect far greater amounts of data and improve its ability to study rare physical processes.





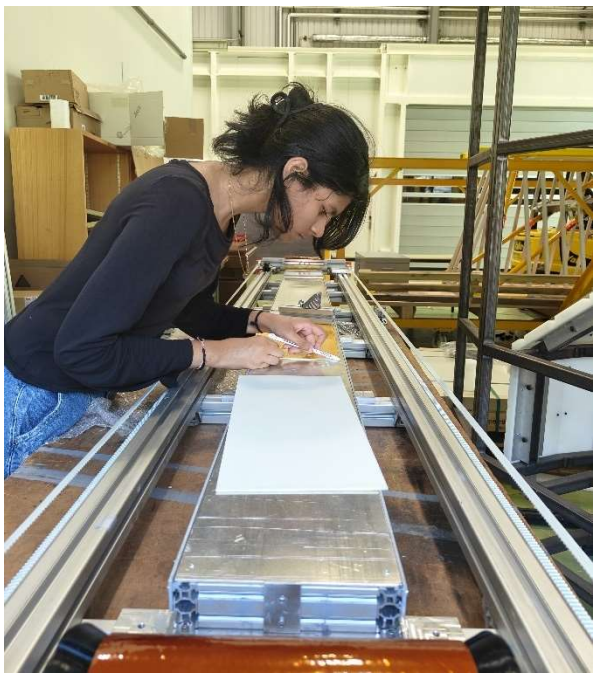




My work involved assembling and preparing detector components, including mounting sensors and preparing the detector for the installation of subsequent components. The images above show me working on the outermost layer of the CMS detector, preparing the structure for the sensors to be mounted along the tracks.

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I prepared protective padding for the CMS shell to prevent damage to the sensitive sensor components. This involved covering foam slabs with Kapton tape, which can withstand extreme temperatures. Since parts of the CMS detector operate at very low temperatures, this makes Kapton tape particularly suitable for protecting its internal components.

Working with these components in a cleanroom gave me a practical understanding of the level of precision required in scientific instrumentation. Even seemingly small steps had to be performed carefully and checked repeatedly, since a minor defect or assembly error could affect the performance of the final detector. I became particularly aware of the importance of cleanliness, careful handling, documentation and systematic verification when working with highly sensitive equipment.

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Cloud Chamber Kit: Making Particle Physics Accessible

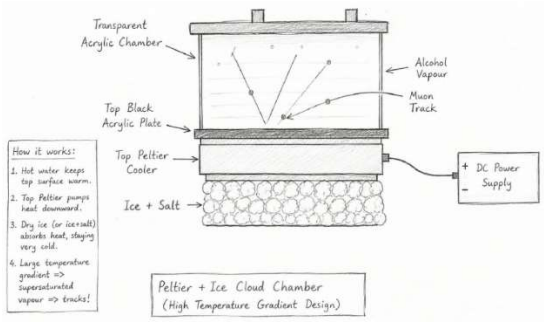
I worked with a team to make particle physics more accessible to Indian school students by developing an affordable cloud chamber kit. Conventional cloud chambers use dry ice to make the paths of charged particles visible, but the cost and difficulty of sourcing and handling dry ice can make them challenging for schools to use.

We investigated alternative cooling methods and explored using reusable gel ice packs instead of dry ice, testing different temperatures to determine the conditions under which particle tracks became visible. We also experimented with cooling the gel packs inside a metal tin. Although we were able to observe particle tracks, they were faint and appeared infrequently.

I also evaluated locally available components to keep the kit's estimated cost at approximately ₹3,000 (USD 30).

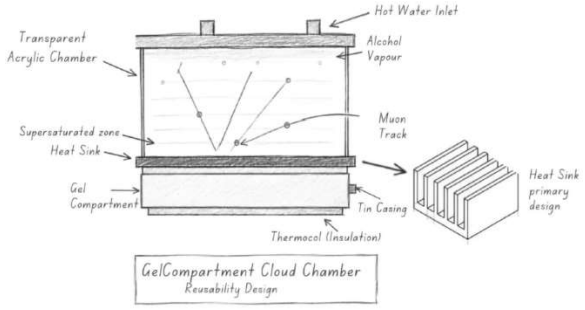
I took lead in designing the kit packaging, organizing the components, providing clear safety instructions, and making the kit easy for students and educators to use.

This experience taught me that a promising design must work under real-world constraints, not just in theory.



Original Proposed Model

Our initial design used a Peltier cooling system, but testing revealed several challenges, particularly the difficulty of dissipating heat from the system and its high cost. Since our primary goal was to develop an affordable cloud chamber, these limitations made the system unsuitable for our needs.



New Proposition

Our final proposed design used a gel-based cooling system. The chamber was placed on a metal tin containing a refrigerated gel pack, which cooled the chamber to the low temperatures needed to make particle tracks visible.

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These experiences showed me two sides of experimental science: developing ways for students to explore the invisible world of particles, and contributing to the laboratory work that makes particle physics research possible.

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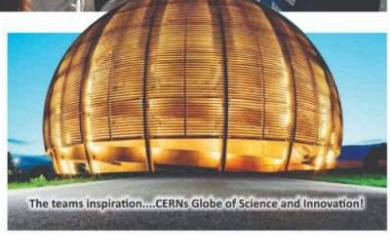
Media Coverage

LEADERS OF CHANGE

Taking Indian science learning to CERN

A team of six Indian students has taken classroom learning to CERN in Geneva, exploring Cloud Chamber technology and gaining hands-on exposure to particle physics, engineering and scientific research





LEARNING PARTICLE PHYSICS AT CERN

A team of bright young Indian students selected from a cohort of international students have been successfully working on a key project at CERN (Geneva, Switzerland), the globally renowned European Organization for Nuclear Research, to study the design, operation and educational applications of the Cloud Chamber, one of the earliest particle detectors capable of making otherwise invisible cosmic particles visible.

The six-member Indian team — Ved Rathi, Arya Rathi, Suchir Divekar, Samyak Mohanty, Pragya Goyal and Sushrut Karmarkar — have been working alongside senior scientists at the world's leading particle physics laboratory. Scientists at CERN have been actively mentoring this team and promoting advanced scientific learning through international research exposure. The initiative aims to bridge the gap between cutting-edge scientific research and classroom education by introducing practical, inquiry-based learning experiences for school students.

EXPLORING THE SCIENCE BEHIND THE CLOUD CHAMBER

During their internship, the students explored the scientific and engineering principles behind the cloud chamber, including thermal management, insulation, particle detection and experimental design. Their work combined concepts of physics, engineering and scientific communication while examining how such technology can be adapted for educational use in schools. The project envisioned by the LifeLab Foundation aims to provide students with hands-on experimental experiences that stimulate curiosity and encourage scientific thinking from an early age instead of limiting the study of atoms, radiation and particle physics only from textbooks.



MENTORSHIP AT THE WORLD'S LEADING PARTICLE PHYSICS LABORATORY

Commending the students' performance, Pieter Everaerts, Senior Physicist with the CMS Experiment at CERN said, "We highly value the curiosity and professionalism of the young Indian students whom the foundation brings to CERN. They have consistently impressed us with their scientific competence, adaptability, and collaborative spirit. It has been a pleasure to mentor such motivated students, whose dedication and strong work ethic speak volumes about their potential. Ultimately, scientific talent thrives wherever young people are provided with opportunity, trust, and supportive mentorship."

The students experienced CERN's collaborative research environment, interacted with leading physicists and engineers, and gained first-hand exposure to the sophisticated technologies used in experiments at the world famous Large Hadron Collider (LHC) and the Globe of Science and Innovation. The experience also highlighted the importance of teamwork, precision, perseverance and problem-solving in scientific research.

A TRANSFORMATIVE EXPERIENCE

Reflecting on the internship, team member Ved Rathi said, "Our internship at CERN is a remarkable experience through which we gained valuable exposure and skills while learning how to solve complex scientific challenges. Exploring the advanced physics labs and equipment strengthened our passion for STEM education. Furthermore, the insightful lectures and the intellectual discussions with senior leadership at CERN further inspired us to pursue research-driven innovation in the future. We would like to thank our mentors at CERN for guiding us throughout the project. Overall, it was an unforgettable experience that broadened both our knowledge and our perspective."



FROM STUDENTS TO SCIENCE AMBASSADORS

There is another important dimension to the students' journey. They should not be seen solely as beneficiaries of education; they can also become its designers, mentors and ambassadors.

"Having studied the Cloud Chamber through direct experience, the students can now help future cohorts understand both the science and the process of experimentation. They can share what worked, what needed to be changed and what they learned from difficulties encountered along the way," says Dr Archana Sharma, Senior Scientist at CERN.

BRINGING CERN'S LEARNING BACK TO INDIAN CLASSROOMS

One of the project's long-term objectives is to establish cloud chamber-based educational programmes in schools across India, supported by trained teachers, classroom resources and practical science modules. This initiative is expected to encourage greater interest in STEM education, inspire future scientists and innovators, and provide young learners an opportunity to witness the invisible world of particle physics through real scientific experimentation.

By connecting Indian classrooms with CERN, one of the world's most advanced scientific institutions, the initiative represents an important step towards nurturing scientific curiosity, strengthening research culture and preparing the next generation of innovators to contribute to global scientific advancement.